

Homework 7

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March 05, 2024

Problem 1.

Proof. Assume the contrary $x < 0$, then let $\epsilon = -x$. By definition of limit, $\exists K \in \mathbb{N} \ni \forall n \geq K, x_n - x < \epsilon \implies x_n < \epsilon + x \implies x_n < 0$, which contradicts the hypothesis. Thus by contradiction, $x > 0$.

$\forall \epsilon > 0$,

The proof is broken down into two cases $x = 0, x \neq 0$

1. **Case 1** $x = 0$: By the definition of a limit, $\exists K \in \mathbb{N} \ni \forall n \geq K, x_n - x < \epsilon^2$. Thus $x_n < \epsilon^2$. Since $x_n > 0$, thus $0 < x_n < \epsilon^2$. Thus $0 < \sqrt{x_n} < \sqrt{\epsilon^2} = \epsilon$, thus $-\epsilon < 0 < \sqrt{x_n} < \epsilon \implies |\sqrt{x_n} - 0| < \epsilon \implies \lim \sqrt{x_n} = 0 = \sqrt{x}$
2. **Case 2** $x \neq 0$: $\forall \epsilon > 0$. By the definition of a limit, $\exists K \in \mathbb{N} \ni \forall n \geq K, |x_n - x| < x\epsilon. \forall n \geq K$.

$$\begin{aligned} |\sqrt{x_n} - \sqrt{x}| &< |\sqrt{x_n} - \sqrt{x}| \frac{|\sqrt{x_n} + \sqrt{x}|}{x} \quad (\text{since } \frac{|\sqrt{x_n} + \sqrt{x}|}{x} > 0) \\ &= \frac{|x_n - x|}{x} \\ &< \epsilon \text{ since } |x_n - x| < x\epsilon \end{aligned}$$

Thus $\lim \sqrt{x_n} = \sqrt{x}$.

Thus $(\sqrt{x_n})$ is convergent to $\sqrt{x}$ □

Problem 2.

Proof. Limit of a nonnegative sequence is nonnegative, $x_n > 0 \implies L \geq 0$.

Let $r \in \mathbb{R}$ be a number such that $0 \leq L < r < 1$. Let $\epsilon = r - L$.

By definition of the limit, $\exists K \in \mathbb{N} \ni \forall n \geq K, \frac{x_{n+1}}{x_n} - L < \epsilon$. Thus $\frac{x_{n+1}}{x_n} < \epsilon + L = r \implies x_{n+1} < rx_n$.
Moreover since $\forall m \in \mathbb{N}, 0 < x_m, 0 < x_{n+1} < rx_n$.

Thus $0 < x_{n+1} < rx_n < r^2 x_{n-1} < \dots < r^{n-K+1} x_K$.

Let $C := x_K r^{-K}$.

Thus $0 < rx_n < Cr^{n+1} \implies 0 < x_n < Cr^n$.

$\forall \epsilon_2 > 0$. Let $M := \frac{1}{r} - 1$, thus $r = \frac{1}{M+1}$. By the archemedian property, $\exists N \in \mathbb{N} \ni N > \frac{1}{M\epsilon} \implies \epsilon > \frac{1}{NM}. \forall n \geq N$

$$\begin{aligned}
|r^n - 0| &= |r^n| = r^n = \frac{1}{(M+1)^n} \\
&= \frac{1}{\sum_{i=0}^n \binom{n}{i} M^i} \text{ (Binomial Expansion)} \\
&< \frac{1}{1 + Mn} \text{ (since } M > 0 \text{ and } 1 > 0 \text{ all terms of the sum are positive thus } (1 + Mn) < (M+1)^n) \\
&< \epsilon \text{ (since } \epsilon > \frac{1}{MN} \text{)}
\end{aligned}$$

Thus $\lim(r_n) = 0$. By Properties of Limit operation, since C is a constant, $\lim(Cr_n) = C \lim(r_n) = 0$.

Since $0 < x_n < Cr^n$, by squeeze theorem $\lim(x_n) = 0$. $\square$

Problem 3.

Proof. Base Case: $1 \leq x_1 = 1 < \sqrt{3} = x_2 < 2$

Induction Hypothesis: Suppose for $n = k$, $1 \leq x_k < x_{k+1} < 2$

Inductive Step: Thus $3 \leq x_k + 2 < x_{k+1} + 2 < 4$ and $1 < \sqrt{3} \leq \sqrt{x_k + 2} = x_{k+1} < \sqrt{x_{k+1} + 2} = x_{k+2} < 2$. Thus $1 \leq x_k < x_{k+1} < 2 \implies 1 \leq x_{k+1} < x_{k+2} < 2$.

Thus by induction, $\forall n \in \mathbb{N} \ 1 \leq x_n < x_{n+1} < 2$.

Thus x_n is bounded and monotonically increasing. By monotonic convergence theorem, x_n converges to a limit.

$X = x_n$ and $X_1 = x_{n+1}$, since the X converges X_1 must converge to the same value by the uniqueness of the limit, since it is the tail of X .

Thus, $\lim x_n = \lim x_{n+1} = \lim \sqrt{x_n + 2}$, By problem 1, $\lim \sqrt{x_n + 2} = \sqrt{\lim x_n + 2}$, by properties of operations on limits, $\lim x_n + 2 = \lim x_n + \lim 2$ limit of a constant is a constant, thus $\lim x_n + 2 = \lim x_n + 2$. Thus, $\lim x_n = \sqrt{\lim x_n + 2}$. Let $L = \lim x_n$, thus $L = \sqrt{L^2 + 2} \implies L^2 - L - 2 = 0$ and since $x_n \geq 1 > 0$, $L > 0$, thus $L = 2$. Thus $\lim x_n = 2$. $\square$

Problem 4.

Proof. $\forall n \in \mathbb{N}$, $y_{n+1} - y_n = \sum_{k=1}^{n+1} \frac{1}{n-k+1} - \sum_{k=1}^n \frac{1}{n+k} = -\frac{1}{n+1} + \frac{1}{2n+1} + \frac{1}{2n+2} = \frac{1}{2n+1} - \frac{1}{2n+2}$, since $2n+1 < 2n+2$, $\frac{1}{2n+1} > \frac{1}{2n+2} \implies y_{n+1} - y_n = \frac{1}{2n+1} - \frac{1}{2n+2} > 0 \implies y_{n+1} > y_n$. Thus y_n is monotonically increasing.

$\forall n \in \mathbb{N}$, $\forall k \in [1, \dots, n]$, $n+k > n \implies \frac{1}{n+k} < \frac{1}{n}$. Thus $y_n = \sum_{k=1}^n \frac{1}{n+k} < \sum_{k=1}^n \frac{1}{n} = \frac{n}{n} = 1$. Thus y_n is bounded above by 1.

Since y_n is monotonically increasing and bounded above by 1, by monotonic convergence theorem it is convergent. $\square$

Problem 5a.

Proof. $a_n = (1 - (-1)^n + \frac{1}{n})$. Let $\epsilon = 1$. $\forall N \in \mathbb{N}$, if N is even let $n = N + 1$, if N is odd, let $n = N$. Let $m = 2n$.

$$\begin{aligned}
 |a_m - a_n| &= \left| 1 - (-1)^{2n} + \frac{1}{2n} - 1 + (-1)^n - \frac{1}{n} \right| \\
 &= \left| (-1)^n - (-1)^{2n} + \frac{1}{2n} - \frac{1}{n} \right| \\
 &= \left| -1 - 1 - \frac{1}{2n} \right| \\
 &= \left| -2 - \frac{1}{2n} \right| \\
 &= 2 + \frac{1}{2n} \\
 &> \epsilon \text{ since } 2n > 0 \text{ and thus } 2 + \frac{1}{2n} > 0
 \end{aligned}$$

Thus a_n is not cauchy. By cauchy convergence theorem, and thus divergent (not convergent). $\square$

Problem 5b.

Proof. $a_n = \sin(\frac{n\pi}{4})$

Let $\epsilon = 1$. $\forall N \in \mathbb{N}$.

Let $n = N + 6 - (N \bmod 8)$ and let $m = N + 10 - (N \bmod 8)$. Thus $n \bmod 8 = 6$ and $m \bmod 8 = 2$ thus $a_n = \sin(\frac{(N+6-N \bmod 8)\pi}{4}) = \sin(6\pi/4) = -1$ and $a_m = \sin(\frac{(N+10-N \bmod 8)\pi}{4}) = \sin(10\pi/4) = 1$.

$$\begin{aligned}
 |a_m - a_n| &= |1 - (-1)| \\
 &= |2| = 2 > \epsilon
 \end{aligned}$$

Thus a_n is not cauchy. By cauchy convergence theorem, and thus divergent (not convergent). $\square$

Problem 6a.

Proof. $\forall \epsilon > 0$. By the Archimedean Property, $\exists K \in \mathbb{N} \ni K > \frac{1}{\epsilon} \implies \epsilon > \frac{1}{K}$. $\forall n \geq K$,

$$\begin{aligned}
 |x_m - x_n| &= \left| \frac{m+1}{m} - \frac{n+1}{n} \right| = \left| \frac{n-m}{mn} \right| \\
 &= \frac{m-n}{nm} \\
 &= \frac{1}{n} - \frac{1}{m}
 \end{aligned}$$

$\frac{1}{n} - \frac{1}{m} - \frac{1}{n} = -\frac{1}{m} < 0$, since $m > 0$. Thus $\frac{1}{n} - \frac{1}{m} < \frac{1}{n}$.
Thus,

$$\begin{aligned}
|x_m - x_n| &< \frac{1}{n} \\
&< \frac{1}{K} \text{ (since } n \geq K) \\
&< \epsilon
\end{aligned}$$

Thus $x_n = \frac{n+1}{n}$ is Cauchy. □

Problem 6b.

Proof.

$$\begin{aligned}
|y_m - y_n| &= \left| \sum_{k=1}^m \left(\frac{1}{2}\right)^k - \sum_{k=1}^n \left(\frac{1}{2}\right)^k \right| = \left| \sum_{k=n+1}^m \left(\frac{1}{2}\right)^k \right| \\
&= \sum_{k=n+1}^m \left(\frac{1}{2}\right)^k \\
&< \sum_{k=n+1}^m \left(\frac{1}{2}\right)^{n+1} \\
&= \frac{m - n}{2^{n+1}}
\end{aligned}$$

□

Problem 6b.

Proof. Let $r = \frac{1}{2}$.

$\forall n \in \mathbb{N}$, $y_n = \sum_{i=1}^n r^i$, $ry_n = r \sum_{i=1}^n r^i = \sum_{i=2}^{n+1} r^i$. $y_n(1 - r) = y_n - ry_n = r - r^{n+1}$. Thus $y_n = \frac{r - r^{n+1}}{1 - r} = \frac{\frac{1}{2} - \frac{1}{2}^{n+1}}{\frac{1}{2}} = 1 - \left(\frac{1}{2}\right)^n$. Thus $y_n = 1 - \left(\frac{1}{2}\right)^n$

Claim: $\forall n \in \mathbb{N}$, $r^n < 1$.

Base Case: $n = 1$, $r^n = \frac{1}{2} < 1$

Induction Hypothesis: Suppose $r^k < 1$

Inductive Step: $rr^k < r^{k+1} < r^k \implies r^{k+1} < 1$.

Thus by induction, $\forall n \in \mathbb{N}$, $r^n < 1$.

$\forall N \in \mathbb{N}$, **Claim:** $\forall n \geq N$, $r^n < r^N$.

Base Case: $n = 1$, $r^{N+1} - r^N = r^N(r - 1) < 0$ since $\frac{1}{2} < 1$.

Induction Hypothesis: Suppose $r^{N+k} < r^N$

Inductive Step: $rr^{N+k} = r^{k+1+N} < r^{N+1} \implies r^{k+1} < r^N$.

Thus by induction, $\forall n \geq N$, $r^n < r^N$.

$\forall \epsilon > 0$. For $\epsilon > 1$, let $K = 0$, thus $\left(\frac{1}{2}\right)^K = 1 < \epsilon$. For $\epsilon < 1$, let $K = \lceil -\log_2(\epsilon) \rceil$. Since $\lceil -\log_2(\epsilon) \rceil > -\log_2(\epsilon)$, $2^{\lceil -\log_2(\epsilon) \rceil} > 2^{-\log_2(\epsilon)} \implies 2^{-\lceil -\log_2(\epsilon) \rceil} < 2^{\log_2(\epsilon)}$

Thus $\left(\frac{1}{2}\right)^K = \left(\frac{1}{2}\right)^{\lceil -\log_2(\epsilon) \rceil} < \left(\frac{1}{2}\right)^{-\log_2(\epsilon)} = \epsilon$. Thus $\left(\frac{1}{2}\right)^K < \epsilon$. $\forall m, n \geq K \ni m > n$.

$$\begin{aligned}
|y_m - y_n| &= \left| \left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^m \right| = \left| \left(\frac{1}{2}\right)^n \left(1 - \left(\frac{1}{2}\right)^{n-m}\right) \right| \\
&= \left(\frac{1}{2}\right)^n \left(1 - \left(\frac{1}{2}\right)^{n-m}\right) \\
&< \left(\frac{1}{2}\right)^n \quad (\text{since } r^{n-m} < 1) \\
&< \left(\frac{1}{2}\right)^K \quad (\text{since } n \geq N \implies r^n < r^K) \\
&< \epsilon
\end{aligned}$$

□

Problem 7.

Proof. $\forall \epsilon > 0$. By the archimedean property, $\exists K \in \mathbb{N} \ni K > \frac{1}{4\epsilon^2} \implies \epsilon > \frac{1}{2\sqrt{K}}$

$$\begin{aligned}
||x_{n+1} - x_n| - 0| &= |\sqrt{n+1} - \sqrt{n}| \\
&= |\sqrt{n+1} - \sqrt{n}| \frac{|\sqrt{n+1} + \sqrt{n}|}{|\sqrt{n+1} + \sqrt{n}|} \\
&= \frac{|1|}{|\sqrt{n+1} + \sqrt{n}|} \\
&= \frac{1}{\sqrt{n+1} + \sqrt{n}} \quad (\text{since } \sqrt{n+1} > 0 \text{ and } \sqrt{n} > 0) \\
&< \frac{1}{2\sqrt{n}} \quad (\text{since } \sqrt{n+1} > \sqrt{n}) \\
&< \frac{1}{2\sqrt{K}} \\
&< \epsilon \quad (\text{since } \epsilon > \frac{1}{2\sqrt{K}})
\end{aligned}$$

Thus, $\lim |x_{n+1} - x_n| = 0$.

Assume the contrary, $\exists n \in \mathbb{N} \ni \sqrt{n} < 1$, thus $\sqrt{n}\sqrt{n} = n < \sqrt{n} \implies n < 1$. This is a contradiction, because $\forall n \in \mathbb{N}, n \geq 1$. Thus by contradiction, $\sqrt{n} \geq 1$

Let $\epsilon = \frac{1}{2}$. $\forall M \in \mathbb{N}, n = M$ and $m = 3M$.

$$\begin{aligned}
|x_m - x_n| &= |\sqrt{m} - \sqrt{n}| = \left| \frac{m - n}{\sqrt{m} + \sqrt{n}} \right| \\
&= \frac{m - n}{\sqrt{m} + \sqrt{n}} \quad (\text{since } m > n \text{ and } \sqrt{m} + \sqrt{n} > 0) \\
&< \frac{m - n}{2\sqrt{n}} \quad (\text{since } m > n \text{ so } \sqrt{m} > \sqrt{n}) \\
&= \frac{3M - M}{2\sqrt{M}} \\
&= \sqrt{M} \\
&\geq 1 > \frac{1}{2} = \epsilon
\end{aligned}$$

Thus x_n is not cauchy.

□